acceptance by many scientists.

“Evolution is a tightly coupled dance, with life and the material environment as partners. From the dance emerges the entity Gaia.”

James Lovelock

Keeping the balance

The hint of mysticism in the Gaia principle chimed with the “New Age” thinking of the time. This helped popularize the idea, but it also led to a negative reception from the scientific establishment. However, behind the Earth “goddess” metaphor was a serious science-based hypothesis that the interactions of living organisms and their physical surroundings—including the cycles of oxygen, carbon, nitrogen, and sulfur—form a dynamic system that stabilizes the environment.

According to Lovelock, Gaia is controlled by the action of “feedback loops,” which are the checks and balances that compensate for disturbances in the system, bringing it back into equilibrium. To function well, life on Earth depends on a particular balance of variables such as water, temperature, oxygen, acidity, and salinity in its environment. When these are constant, Earth is in a stable state of homeostasis, but if the balance is disturbed, the planet encourages the organisms that will restore the equilibrium, while being hostile to those that reinforce the disturbance. The organic components of the Earth system do not simply react to changes in their environment, but control and regulate them.

These feedback mechanisms operate in a complex global network of interconnected natural cycles, to maintain the optimum conditions for the organisms within them. They can resist change, but only to a certain extent. A big enough disturbance can push the system to a “tipping point,” where, with the balance of its components altered, it is likely to settle into a very different state of equilibrium. Such a tipping point, argued Lovelock,